

Data Representation, Number Systems & Logic

Lesson 50 — From Binary Digits to Gates

JKSSB Computer Awareness

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Why Computers Use Binary

- ▶ Digital circuits can reliably distinguish two states, commonly labelled 0 and 1.
- ▶ A **number system** uses a base (radix) and a set of allowed digits.
- ▶ In base b , each position represents a power of b .

Positional value

$$1011_2 = 1(2^3) + 0(2^2) + 1(2^1) + 1(2^0) = 11_{10}.$$

Four Number Systems

Name	Base	Digits
Binary	2	0–1
Octal	8	0–7
Decimal	10	0–9
Hexadecimal	16	0–9, A–F

In hexadecimal, A=10, B=11, C=12, D=13, E=14, and F=15.

Convert Decimal to Binary

Convert 45_{10} to binary

Divide repeatedly by 2 and read the remainders from bottom to top.

- ▶ $45 \div 2 = 22$ remainder 1
- ▶ $22 \div 2 = 11$ remainder 0; $11 \div 2 = 5$ remainder 1
- ▶ $5 \div 2 = 2$ remainder 1; $2 \div 2 = 1$ remainder 0
- ▶ $1 \div 2 = 0$ remainder 1

Result

$45_{10} = 101101_2$.

Binary, Octal, and Hexadecimal

- ▶ Group binary digits in threes to convert to octal:
 $1111111_2 = 001\ 111\ 111_2 = 177_8$.
- ▶ Group binary digits in fours to convert to hexadecimal: $0111\ 1111_2 = 7F_{16}$.
- ▶ Verify by place value: $7F_{16} = 7(16) + 15 = 127_{10}$.

Pad the leftmost group with zeros when needed; do not change the value.

Text and Decimal Digits Have Encodings

- ▶ **ASCII**: character-encoding standard; original ASCII uses 7-bit codes.
- ▶ **Unicode**: assigns code points to characters across many writing systems; UTF-8 is a variable-length encoding.
- ▶ **BCD**: Binary-Coded Decimal; each decimal digit is represented by a 4-bit code.
- ▶ These are not interchangeable: a character code, a Unicode encoding, and a decimal-digit representation solve different problems.

Representing Colour: RGB

- ▶ RGB represents colour with red, green, and blue channel values.
- ▶ In a common 24-bit RGB representation, each channel has 8 bits: $8 + 8 + 8 = 24$ bits per pixel.
- ▶ More bits per channel can represent more levels; actual image formats and display systems may use other layouts.

Keep the claim scoped

24-bit RGB is a common representation, not a rule for every image or display.

Basic Logic Gates

A	B	AND	OR	XOR	NAND
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	1
1	1	1	1	0	0

- ▶ AND is 1 only when both inputs are 1.
- ▶ OR is 1 when at least one input is 1.
- ▶ XOR is 1 when the inputs differ; NAND is the negation of AND.

NOT Gate and Exam Traps

- ▶ NOT has one input and reverses it: NOT 0 = 1; NOT 1 = 0.
- ▶ Do not call BCD ordinary binary for the whole number: BCD encodes each decimal digit separately.
- ▶ Do not call UTF-8 fixed-length: it is variable-length.
- ▶ Do not confuse RGB bits per pixel with the number of colour channels.

Lesson 50: Recall

- ▶ Binary, octal, decimal, and hexadecimal use bases 2, 8, 10, and 16.
- ▶ Place values are powers of the base.
- ▶ ASCII, Unicode/UTF-8, BCD, and RGB represent different kinds of information.
- ▶ Truth tables give the output for every input combination.